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Thermodynamic optimization of the Ni–Al–Nd ternary system

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ABSTRACT

Thermodynamic optimization of the Ni-Al-Nd ternary system and Al-Nd binary systems have been conducted in the present work. A self-consistent set of thermodynamic parameters for the Al-Nd binary system and Ni-Al-Nd ternary system have been optimized using CALPHAD method. Isothermal sections at 600 and 700 °C as well as the liquidus projection have been reproduced. Isopleths with 93 at% Al, 9 at% Ni and 3 at% Nd, have been calculated also. The calculated thermodynamic and phase equilibria data for both the binary and the ternary systems agree fairly well with the experimental data. This work can be used as multi-component thermodynamic database for Ni-based alloys.

1. Introduction

Single-crystal Ni-based superalloys have an excellent elevated-temperature rupture life and oxidation resistance, which is attributed to highly complex alloying because a large amount of metal elements, such as Cr, Al, W, V and others, are added to the alloy [1–3]. Besides these alloying elements, small amount of rare-earth element i.e. 0–0.1 wt% Nd can refine the microstructure and improve the alloy grain-boundary strength. The element Nd can decrease the surface tension of the liquid and the energy to form grain nuclei leading to more nuclei; in addition, the Nd oxides and sulfides may also act as crystallizing nuclei and prevent the grains from growing [4,5]. To understand the mechanism of the influence of rare metals on superalloys more clearly, a CALPHAD (CALculation of PHAse Diagram) method [6] combined with a combinatorial high-throughput experiment was used to achieve quantitative analyses of the effects of rare metals on superalloys, and the alloy properties can be predicted [7–9]. A thermodynamic database that contains rare-earth element Nd can be useful for the design of Ni-based superalloys. We optimized the Ni–Al–Nd ternary system as a constitutional system in the Pan-Nickel database [10].

2. Literature review

2.1. Ni–Al–Nd ternary system

The Ni–Al–Nd ternary-system experimental information has been summarized by Raghavan [11] and assessed by Ferro et al. [12]. Rykhal [13] determined the isothermal section at 600 °C within a range of 0–33.3 at% Nd and reported eight ternary compounds and some of the crystal structures. Some authors [14–16] have studied the crystal structure of ternary compounds in the Ni–Al–Nd, system. Information on the ternary compounds is presented in Table 1. Ni_5Nd compounds exhibit an extended solid solubility of 35 at% Al. The solid solubility of Rykhal for $NdAl_2$–$NdNi_2$ shows a variance from the data reported by Leon and Wallace [17] who claimed a solubility of up to 5 mol% $NdAl_2$ in $NdNi_2$.

The phase equilibria of the Al-rich portion of the Ni–Al–Nd ternary system has been investigated by Gödecek et al. [18,19] using metallography analysis, thermal and magnetothermal analysis and X-ray analysis. Two isothermal sections at an Al-rich corner were given at 600 and 700 °C, respectively, and a ternary eutectic reaction Liq→Fcc $+\alpha_Al_{11}Nd_3+\tau_1$ existed at 626 °C. A small solubility range has been reported for Nd in the Ni sublattice of the τ_1 phase. Isopleths with 93 at % Al, 9 at% Ni and 3 at% Nd, were also studied from DTA data. It is worth noting that Fcc-Al phase is in equilibrium with τ_1 phase rather than Nd_3Al_{11} and $NiAl_3$ phases according to the metallographic analysis results by Gödecek et al. [18,19].

Thermodynamic information of Ni–Al–Nd ternary systems is rare. Enthalpies of formation of the ternary intermediate τ_7 (NiAlNd) were determined by Borzone et al. [20] using high-temperature direct calorimetry. The free first-principles databases (OQMD [21] and Materials Project [22]) used an ab initio method to calculate the enthalpies of formation of some ternary intermediates.

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Table 1
Ternary intermetallic compounds.

Phase		Prototype	Composition (at%)	Source
τ_1	Ni_2Al_7Nd	–	70.00 Al, 10.00 Nd	[13]
τ_2	$NiAl_4Nd$	YNiAl4	66.67 Al, 16.67 Nd	[13]
τ_3	Ni_2Al_3Nd	$CaCu_5$	50.00 Al, 16.67 Nd	[13]
τ_4	$Ni_{2.7}Al_{2.3}Nd$	–	38.33 Al, 16.67 Nd	[13]
τ_5	$NiAl_2Nd$	$CuMgAl_2$	50.00 Al, 25.00 Nd	[16]
τ_6	Ni_8AlNd_3	Ce_3Co_8Si	8.33 Al, 25.00 Nd	[13]
τ_7	NiAlNd	ZrNiAl	33.33 Al, 33.33 Nd	[13,14]
τ_8	Ni_2AlNd_2	MoNiB	20.00 Al, 40.00 Nd	[13,15]

2.2. Al–Nd binary system

The phase-equilibrium information of an Al–Nd binary system has been investigated by many authors. Buschow [23] first determined the entire composition by metallography, thermal analysis and X-ray diffractometry, and found four solution phases (liquid, fcc, bcc and dhcp) and five intermetallic compounds (Al_4Nd, Al_3Nd, Al_2Nd, AlNd, $AlNd_2$ and $AlNd_3$), including an allotropic transformation α_Al_4Nd/β_Al_4Nd at 950 °C. Later, Buschow published another paper that corrected the phase to $Al_{11}Nd_3$ by using a full-matrix least-squares refinement on the basis of a threefold unit cell of $A1_4Ba$ [24], and the temperature was modified to 934 °C by Gödecke et al. [18] using metallography, differential-thermo analysis, magneto-thermal analysis and X-ray analysis. In 1990, Kononenko and Golubev [25] measured the phase equilibria for the Al-rich corner of the Al–Nd system, and the eutectic reaction Liq→Fcc + α-$Al_{11}Nd_3$ temperature was determined to be 632 °C, which is 8 °C lower than that in [23]. In a recent study, Cao et al. showed that the eutectic temperature is in the order of 645 °C [26]. A Nd-rich corner up to 30 at% Al of the Al–Nd system was re-investigated in 1996 by Saccone et al. [27]. The temperatures of reactions Liq + $AlNd_2$→$AlNd_3$ is approximately 100 °C higher than those measured by Buschow [23]. The slight solubility of Nd in the (Al) solution was determined to be up to 0.008 at% by Drits et al. [28].

The enthalpies of mixing for the liquid at 977 °C and 1227 °C were determined by Zviadadze et al. [29] using high-temperature calorimetry. The Al–Nd binary system contains very stable intermetallic compounds, and their enthalpy of formation has been studied by many groups. Kober et al. [30] determined the enthalpy of formation of the $Al_{11}Nd_3$ phase by the e.m.f. method. The enthalpy of formation of the Al_2Nd phase has been measured by Colinet et al. by using dissolution calorimetry [31]. The direct-synthesis calorimetry method has been used by Borzone et al. [32] to obtain the enthalpies of formation of all intermediate phases. First-principles calculations were conducted by Gao et al. to derive the enthalpies of formation of the intermediate phases [33],the free first-principles databases (OQMD [21] and Materials Project [22]) also provide.

A thermodynamic optimization of the Al–Nd system was conducted by various authors. In the early years, this system was optimized thermodynamically by Wang [34], Clavaguera and Du [35] and Cacciamani and Ferro [36] without taking into account the latest experimental data [18,19]. The modified quasichemical model was applied to the liquid phase in the work by Jin [37], which is not applicable in the Pan–Nickel database. In this work, we have re-optimized the Al–Nd binary system and the solubility of Nd in Al was considered.

2.3. Ni–Nd binary system

The phase-equilibrium information of the Ni–Nd binary system was studied by Pan et al. [39], Qi et al. [40] and Huang et al. [41] three solution phases (fcc, bcc and hcp), a liquid phase and eight intermetallic compounds (Ni_3Nd_7, $NiNd_3$, NiNd, Ni_3Nd_2, $Ni_{16}Nd_7$, Ni_3Nd, Ni_7Nd_2 and Ni_5Nd), which included an allotropic transition α_Ni_7Nd_2/β_Ni_7Nd_2. No solubility was found for Nd in Ni [39]. Many authors have optimized the Ni–Nd system thermodynamically [39,41–43]. Luo's optimization work was consistent with the experimental data and the Ni–Nd phase is as shown in Fig. 1a. Therefore, we have taken Luo's thermodynamic parameters into consideration in this work.

2.4. Ni–Al binary system

The Ni–Al system was investigated extensively. Thermodynamic parameters of the Ni–Al system have been adopted from previous work [38] without changes. The phase diagram is as shown in Fig. 1b.

3. Thermodynamic models

3.1. Solution phase

Pandat software [44] was used. The thermodynamic model for the Gibbs free energy G of the solution phases, including liquid, fcc, bcc and dhcp is expressed as:

$$G_m^{\varphi} = \sum_{i=\mathrm{Al,Ni,Nd}} ({}^0G_i^{\varphi}) + RT \sum_{i=\mathrm{Al,Ni,Nd}} (x_i \ln(x_i)) + {}^{\mathrm{ex}}G_m^{\varphi} \tag{1}$$

in which i represents elements Al, Nd and Ni; x is the composition; ${}^0G_i^{\varphi}$ is the Gibbs free energy for pure elements i in the structure φ phase from Dinsdale [45]; R is the gas constant; T is the temperature in K and ${}^{\mathrm{ex}}G_m^{\varphi}$ is the excess Gibbs free energy that is described as:

$${}^{\mathrm{ex}}G_m^{\varphi} = x_{\mathrm{Al}}x_{\mathrm{Ni}}L_{\mathrm{Al,Ni}}^{\varphi} + x_{\mathrm{Al}}x_{\mathrm{Nd}}L_{\mathrm{Al,Nd}}^{\varphi} + x_{\mathrm{Nd}}x_{\mathrm{Ni}}L_{\mathrm{Nd,Ni}}^{\varphi} + x_{\mathrm{Al}}x_{\mathrm{Ni}}x_{\mathrm{Nd}}L_{\mathrm{Al,Ni,Nd}}^{\varphi} \tag{2}$$

where $L_{\mathrm{Al,Ni}}^{\varphi}$, $L_{\mathrm{Al,Nd}}^{\varphi}$ and $L_{\mathrm{Nd,Ni}}^{\varphi}$ are the interaction parameters in the Al–Ni, Al–Nd and Nd–Ni binary system and $L_{\mathrm{Al,Ni,Nd}}^{\varphi}$ corresponds to the interaction parameters of the Ni–Al–Nd ternary system. All parameters can be obtained according to optimization using experimental data.

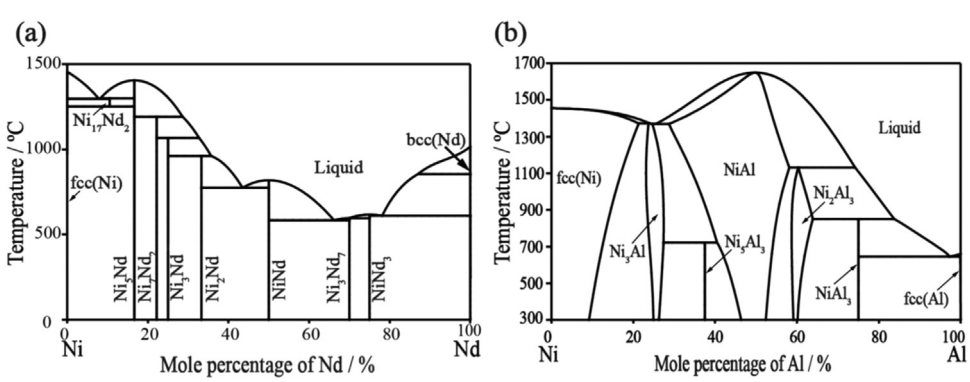

Fig. 1. Calculated Ni–Nd phase diagram (a) and calculated Ni–Al phase diagram (b).

Table 2
Optimized thermodynamic parameters in this work.

Phase		Thermodynamic parameters
	Liquid	${}^{0}L^{\mathrm{Liq}}_{\mathrm{Al,Nd}} = -128300 + 24T,\ {}^{1}L^{\mathrm{Liq}}_{\mathrm{Al,Nd}} = -26790 + 0.5,\ {}^{2}L^{\mathrm{Liq}}_{\mathrm{Al,Nd}} = -8500$
		${}^{0}L^{\mathrm{Liq}}_{\mathrm{Al,Ni:Nd}} = -40573 - 58T$
	fcc	${}^{0}L^{\mathrm{fcc}}_{\mathrm{Al,Nd}} = -73434\ ,\ {}^{1}L^{\mathrm{fcc}}_{\mathrm{Al,Nd}} = -13134$
	bcc	${}^{0}L^{\mathrm{bcc}}_{\mathrm{Al,Nd}} = -99870 + 13T,\ {}^{1}L^{\mathrm{bcc}}_{\mathrm{Al,Nd}} = -7500$
	dhcp	$L^{\mathrm{dhcp}}_{\mathrm{Al,Nd}} = -67800$
	Ni_5Nd	${}^{0}G^{\mathrm{Ni_5Nd}}_{\mathrm{Al:Nd}} = {}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + 5{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} + 244800,\ L^{\mathrm{Ni_5Nd}}_{\mathrm{Al,Ni:Nd}} = -285000 + 1T$
	$\alpha_Al_{11}Nd_3$	${}^{0}G^{\alpha_\mathrm{Al_{11}Nd_3}}_{\mathrm{Al:Nd}} = 3{}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + 11{}^{0}G^{\mathrm{fcc}}_{\mathrm{Al}} - 532011 + 66.79T$
	$\beta_Al_{11}Nd_3$	${}^{0}G^{\beta_\mathrm{Al_{11}Nd_3}}_{\mathrm{Al:Nd}} = 3{}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + 11\ {}^{0}G^{\mathrm{fcc}}_{\mathrm{Al}} - 530804 + 65.79T$
	Al_3Nd	${}^{0}G^{\mathrm{Al_3Nd}}_{\mathrm{Al:Nd}} = {}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + 3\ {}^{0}G^{\mathrm{fcc}}_{\mathrm{Al}} - 175452 + 27.3325T$
	Al_2Nd	${}^{0}G^{\mathrm{Al_2Nd}}_{\mathrm{Al:Nd}} = {}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + 2\ {}^{0}G^{\mathrm{fcc}}_{\mathrm{Al}} - 162511 + 28.18T\ {}^{0}G^{\mathrm{Al_2Nd}}_{\mathrm{Ni:Al}} = {}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + 2\ {}^{0}G^{\mathrm{fcc}}_{\mathrm{Al}} + 45000162511 + 28.18T$
		$45000\ L^{\mathrm{Al_2Nd}}_{\mathrm{Al,Ni:Nd}} = -274500\ L^{\mathrm{Al_2Nd}}_{\mathrm{Nd:Al,Ni}} = 24000$
	AlNd	${}^{0}G^{\mathrm{Al\ Nd}}_{\mathrm{Al:Nd}} = {}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + {}^{0}G^{\mathrm{fcc}}_{\mathrm{Al}} - 94720 + 19.88T$
	$AlNd_2$	${}^{0}G^{\mathrm{AlNd_2}}_{\mathrm{Al:Nd}} = 2{}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + {}^{0}G^{\mathrm{fcc}}_{\mathrm{Al}} - 100000 + 18.56T$
	$AlNd_3$	${}^{0}G^{\mathrm{AlNd_3}}_{\mathrm{Al:Nd}} = 3{}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + {}^{0}G^{\mathrm{fcc}}_{\mathrm{Al}} - 101911 + 15.47T$
τ_1	Ni_2Al_7Nd	${}^{0}G^{\mathrm{Ni_2Al_7Nd}}_{\mathrm{Ni:Al:Nd}} = 7{}^{0}G^{\mathrm{fcc}}_{\mathrm{Al}} + {}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + 2{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 480841 + 35T$
		$L^{\mathrm{Ni_2Al_7Nd}}_{\mathrm{Ni,Nd:Al:Nd}} = -267800 + 15{*}T$
τ_2	$NiAl_4Nd$	${}^{0}G^{\mathrm{Ni\ Al_4Nd}}_{\mathrm{Ni:Al:Nd}} = 4{}^{0}G^{\mathrm{fcc}}_{\mathrm{Al}} + {}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 345736 + 50.4T$
τ_3	Ni_2Al_3Nd	${}^{0}G^{\mathrm{Ni_2Al_3Nd}}_{\mathrm{Ni:Al:Nd}} = 3{}^{0}G^{\mathrm{fcc}}_{\mathrm{Al}} + {}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + 2{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 398482 + 1.5T$
τ_4	$Ni_{2.7}Al_{2.3}Nd$	${}^{0}G^{\mathrm{Ni_{2.7}Al_{2.3}Nd}}_{\mathrm{Ni:Al:Nd}} = 2.3{}^{0}G^{\mathrm{fcc}}_{\mathrm{Al}} + {}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + 2.7{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 417500$
τ_5	$NiAl_2Nd$	${}^{0}G^{\mathrm{Ni\ Al_2Nd}}_{\mathrm{Ni:Al:Nd}} = 2{}^{0}G^{\mathrm{fcc}}_{\mathrm{Al}} + {}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 258850 + 3T$
τ_6	Ni_8AlNd_3	${}^{0}G^{\mathrm{Ni_8Al\ Nd_3}}_{\mathrm{Ni:Al:Nd}} = {}^{0}G^{\mathrm{fcc}}_{\mathrm{Al}} + 3{}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + 8{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 520000 - 78T$
τ_7	NiAlNd	${}^{0}G^{\mathrm{Ni\ Al\ Nd}}_{\mathrm{Ni:Al:Nd}} = {}^{0}G^{\mathrm{fcc}}_{\mathrm{Al}} + {}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 171500 - 33T$
τ_8	Ni_2AlNd_2	${}^{0}G^{\mathrm{Ni_2Al\ Nd_2}}_{\mathrm{Ni:Al:Nd}} = 2{}^{0}G^{\mathrm{fcc}}_{\mathrm{Al}} + {}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} + 2{}^{0}G^{\mathrm{fcc}}_{\mathrm{Ni}} - 275085 + 2T$

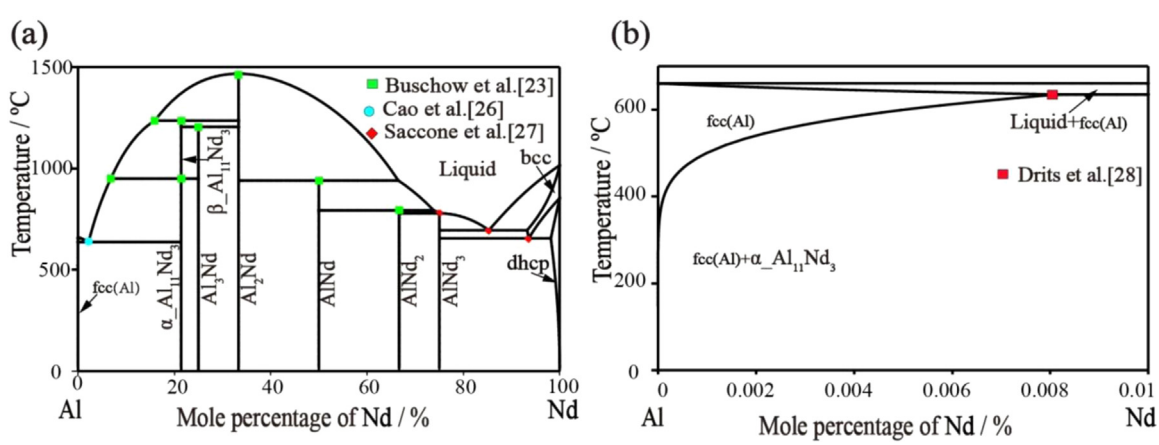

Fig. 2. Calculated Al–Nd phase diagram. (a) Full composition range; (b) magnified Al-rich corner showing solubility of Nd in Al.

3.2. Binary compounds

Thermodynamic models for most of the binary phases are taken from previous work, except for the Ni_5Nd phase because of the solubility of the Ni_5Nd and Al_2Nd compounds, so the thermodynamic model for the Ni_5Nd and Al_2Nd phases has been modified to:

$$G^{(\mathrm{Al,Ni})_m\mathrm{Nd}_n} = y_{\mathrm{Al}}G_{\mathrm{Al:Nd}} + y_{\mathrm{Ni}}G_{\mathrm{Ni:Nd}} + mRT(y_{\mathrm{Al}}\ln(y_{\mathrm{Al}}) + y_{\mathrm{Ni}}\ln(y_{\mathrm{Ni}})) + y_{\mathrm{Al}}y_{\mathrm{Ni}}L_{\mathrm{Nd:Al,Ni}} \quad (3)$$

in which y_{Al} and y_{Ni} are the site fractions of Al and Ni in the first sublattice, m and n are stoichiometric numbers, $G_{\mathrm{Al:Nd}}$ and $G_{\mathrm{Nd:Ni}}$ are the Gibbs free energies for end-members and $L_{\mathrm{Nd:Al,Ni}}$ is the interaction term between the two sublattices to be optimized.

3.3. Ternary compounds

Eight ternary intermetallic compounds were considered. Most ternary intermetallic compounds were treated as a $Ni_mAl_nNd_l$ stoichiometric compound model. The Gibbs free energy is:

$$G^{\mathrm{Ni_mNi_nNi_l}}_{\mathrm{Al:Ni:Nd}} = nx_{\mathrm{Al}}G^{\mathrm{fcc}}_{\mathrm{Al}} + lx_{\mathrm{Nd}}{}^{\mathrm{dhcp}}_{\mathrm{Nd}} + x_{\mathrm{Ni}}{}^{\mathrm{fcc}}_{\mathrm{Ni}} + A + BT \quad (4)$$

where A and B are the thermodynamic parameters to be optimized. The τ_1 phase (Ni_2Al_7Nd) has the solubility range, and so the Gibbs free energy can be described as:

Table 3
Invariant reactions in the Al–Nd binary system.

Invariant reaction	Temperature, °C		Composition% $X_{(Nd)}$		
	Calculated	Experiment	First phase	Second phase	Third phase
Liq →fcc + α_$Al_{11}Nd_3$	635	640 [18]	2.31	0	21.42
Liq + Al_2Nd→β_$Al_{11}Nd_3$	1235	1235[11]	17.62	33.33	21.42
β_$Al_{11}Nd_3$ + Al_2Nd →Al_3Nd	1204	1205[11]	21.42	33.33	25
Liq + Al_2Nd→AlNd	945	940 [11]	63.95	33.33	50.00
β_$Al_{11}Nd_3$→Al_3Nd + α_$Al_{11}Nd_3$	934	934[18]	21.42	0.25	21.42
Liq + β_$Al_{11}Nd_3$→α_$Al_{11}Nd_3$	934	934 [18]	6.88	21.42	21.42
Liq + AlNd→$AlNd_2$	799	795[13]	72.21	50.00	66.67
Liq + $AlNd_2$→$AlNd_3$	776	780 [13]	75.01	33.3	75.00
Liq →$AlNd_3$ + bcc	690	1240 [11]	85.54	75	–
bcc→$AlNd_3$ + dhcp	650	873 [13]	93.27	25	96.63

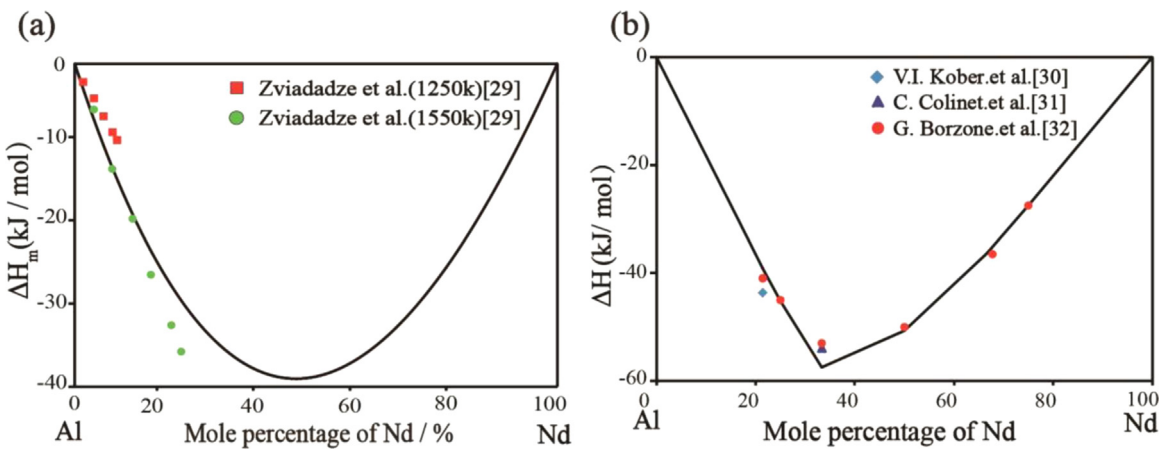

Fig. 3. Calculated enthalpies of mixing of the Al–Nd liquid phase (a) and calculated enthalpies of formation of intermediate phases (b).

Table 4
Calculated enthalpies of formation of ternary intermetallic compounds compared with other sources.

Phase		Enthalpy of formation/kJ mol^{-1}			
		Measured	OQMD	Materials project	Present work
τ_1	Ni_2Al_7Nd	···	···	···	–48.08
τ_2	$NiAl_4Nd$	···	–52.97		–57.62
τ_3	Ni_2Al_3Nd	···	–59.34		–66.41
τ_4	$Ni_{2.7}Al_{2.3}Nd$	···	···	···	–38.00
τ_5	$NiAl_2Nd$	···	–14.47		–64.71
τ_6	Ni_8AlNd_3	···	–33.28		–43.33
τ_7	NiAlNd	–55 ± 2	–49.98	–49.30	–57
τ_8	Ni_2AlNd_2	···	–42.84	–43.03	–55.01

$$G^{Al_7(Ni,Nd)_2Nd} = y_{Al}y_{Ni}G_{Al:Ni:Nd} + y_{Al}y_{Nd}G_{Al:Nd:Nd} + 2RT(y_{Ni}\ln(y_{Ni}) + y_{Nd}\ln(y_{Nd})) + y_{Al}y_{Ni}y_{Nd}L_{Al:Ni,Nd:Nd} \quad (5)$$

in which y_{Nd} and y_{Ni} are the site fractions of Nd and Ni in the first sublattice, $G_{Al:Ni:Nd}$, $L_{Nd:Al,Ni}$ and $G_{Al:Nd:Nd}$ is the interaction term to be optimized.

4. Results and discussion

4.1. Binary system

Calculated phase diagrams for the Ni–Nd and Ni–Al binary systems are presented in Fig. 1a and Fig. 1b. For the Al–Nd binary system, the optimized thermodynamic parameters are listed in Table 2, and the optimized Al–Nd phase diagram is shown in Fig. 2a. Compared with the literature data, the solubility of Nd in Al is considered to be ~0.008 at

Table 5
Calculated invariant reaction of the Ni–Al–Nd ternary system.

No.	Invariant reaction	Temperature. °C	Composition, %		
			Ni	Al	Nd
P1	Liq + τ_7 + Ni_5Nd→τ_3	1816	30.3	62.3	7.4
P2	Liq + τ_7 + τ_3→ τ_5	1515	14.0	69.4	16.6
U1	Liq + Ni_5Nd → τ_3 + NiAl	1468	36.2	62.2	1.6
U2	Liq + τ_7 → τ_5 + Al_2Nd	1415	5.9	67.2	26.9
U3	Liq + fcc→ Ni_5Nd + Ni_3Al	1371	75.8	24.1	0.1
E1	Liq → Ni_5Nd + NiAl Ni_3Al	1368	74.5	25.5	0.1
U4	Liq + Al_2Nd→τ_5 + + β_$Al_{11}Nd_3$	1158	7.3	77.6	15.1
U5	Liq + τ_5→τ_3 + + β_$Al_{11}Nd_3$	1122	8.6	78.1	13.3
U6	Liq + NiAl→τ_3 + Ni_2Al_3	1122	25.0	73.4	1.6
P3	Liq + τ_7 →τ_3 + + β_$Al_{11}Nd_3$	1053	9.7	80.0	10.3
U7	Liq + τ_3→τ_1 + + β_$Al_{11}Nd_3$	1053	9.7	80.0	10.3
U8	Liq + τ_2→τ_1 + + β_$Al_{11}Nd_3$	1052	9.7	80.0	10.3
U9	Liq + Ni_5Nd → τ_7 + τ_6	1025	42.1	1.6	56.3
U10	Liq + τ_3→τ_1 + Ni_2Al_3	1001	20.5	77.1	2.3
P4	Liq + τ_1 + + β_$Al_{11}Nd_3$ →α_$Al_{11}Nd_3$	934	6.4	87.0	6.6
U11	Liq + Al_2Nd →τ_7 + AlNd	910	1.0	33.9	65.1
U12	Liq + Ni_2Al_3 →τ_1 + $NiAl_3$	860	16.2	83.0	0.8
U13	Liq + bcc →τ_7 + dhcp	818	9.1	6.6	84.3
U14	Liq + AlNd →τ_7 + $AlNd_2$	794	0.9	27.8	71.3
U15	Liq + $AlNd_2$→τ_7 + $AlNd_3$	769	0.9	25.0	74.1
U16	Liq + τ_6→NiNd + τ_7	692	38.2	1.0	60.8
E2	Liq→τ_7 + bcc + NiNd	676	1.4	14.5	84.1
U17	Liq + $NiAl_3$→fcc + τ_1	635	1.8	96.7	1.5
E3	Liq→τ_1 + + fcc + α_$Al_{11}Nd_3$	629	1.0	96.8	2.2
P5	Liq + NiNd + τ_7→τ_8	626	35.3	1.1	63.6
E4	Liq→τ_7 + $NiNd_3$ + Ni_3Nd_7	586	29.3	1.7	68.9
U18	Liq + τ_7→ Ni_3Nd_7 + + τ_8	581	32.9	1.2	65.9
E5	Liq→τ_7 + dhcp + $NiNd_3$	579	20.6	3.7	75.7
E6	Liq→τ_8 + Ni_3Nd_7 + NiNd	578	33.4	1.1	65.5

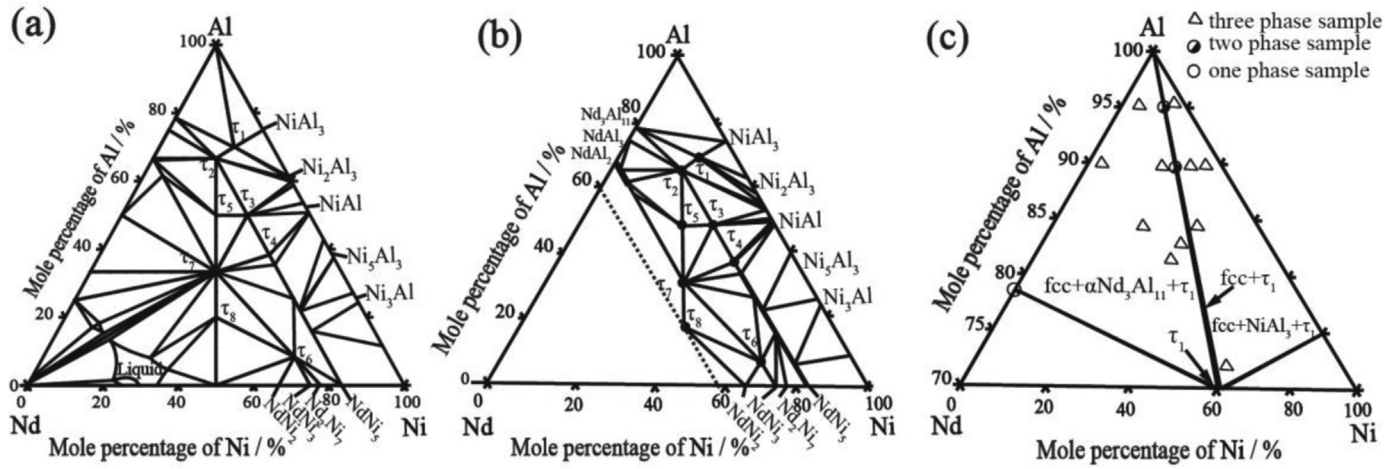

Fig. 4. Calculated isothermal section of the Ni–Al–Nd ternary system at 600 °C compared with the experimental results. (a) Calculated, (b) experimental, [19] (c) experimental. [13].

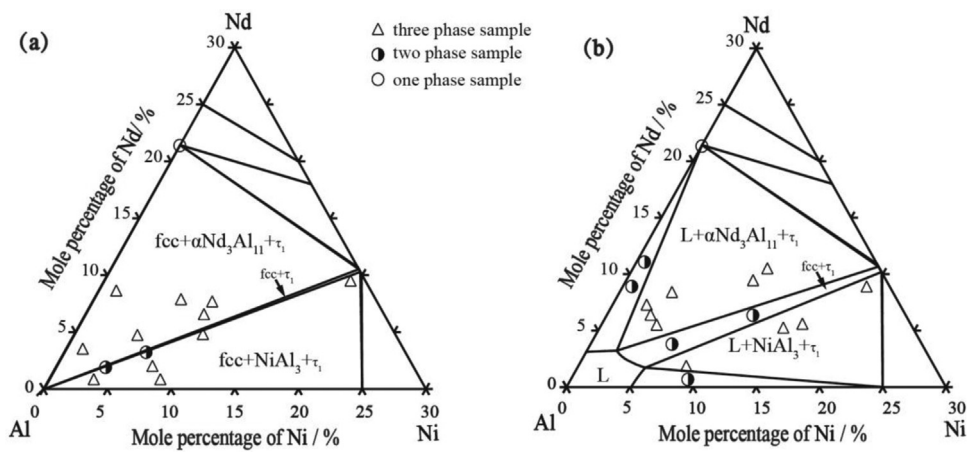

Fig. 5. Calculated isothermal section of the Ni–Al–Nd ternary system at 600 °C (a) and 700 °C (b) compared with experimental results [19].

%, which is consistent with the experimental data in Fig. 2b. All invariant reactions are presented in Table 3 in contrast with the literature. In general, a good agreement is obtained between the calculated phase-equilibrium result and the experimental data.

The calculated enthalpy and entropy of mixing of the liquid phase in the Al–Nd binary system is shown in Fig. 3a compared with the experimental data [29]. Standard enthalpies of formation of the intermetallic compounds in the Al–Nd binary system are calculated and plotted in Fig. 3b for comparison with the measured data [14,17–22] and free first-principles databases (OQMD and Materials Project). Calculated values of thermodynamic properties are consistent with the experimental data, and the entropy values (higher than −10 J/mol K) of mixing in this work are reasonable. Overall, the calculated phase diagram can reproduce the literature data well.

4.2. Ternary system

Optimized thermodynamic parameters in the Ni–Al–Nd ternary system are listed in Table 2. The enthalpy of formation of the ternary compounds has been calculated and compared with different sources in Table 4. The enthalpy of formation is consistent with the measured data, [20] for the τ_7 phase. For other ternary compounds, the calculated enthalpies of formation that were obtained by using the CALPHAD method were lower than the calculated values obtained from the OQMD and Materials Projects database, which used the ab initio method (Table 5).

The phase-equilibrium information at 600 °C as calculated by using these parameters is shown in Fig. 4a and compared with the experimental data in Fig. 4b [13]. The isothermal section of the Al-rich corner at 600 and 700 °C was calculated as shown in Fig. 5a and Fig. 5b, respectively, and compared with the experimental data measured by [18]. Three vertical sections (93 at% Al, 9 at% Ni and 3 at% Nd) were also presented in Fig. 6a–c. The experimental data were marked by using different symbols for the phase region. Our calculated phase equilibria fit the measured data well.

A liquidus projection has been calculated with isothermal lines shown in Fig. 7 with a magnified Al-rich corner (20%). The calculated eutectic reaction Liq→τ_1 + + fcc + α_$Al_{11}Nd_3$ temperature was 629 °C and Liq + $NiAl_3$→fcc + τ_1 was 633 °C, which is slightly higher than the experimental value (626 and 629 °C). Overall, our calculated resulted can reproduce the experimental data well.

5. Conclusions

A Ni–Al–Nd ternary system has been assessed by using the CALPHAD approach. Eight ternary intermetallic compounds were considered and optimized. The ternary solubility of the Al_2Nd and Ni_5Nd have also been considered. The calculated phase-equilibria information of the isothermal and vertical section is consistent with the experimental data.

The re-optimized Al–Nd binary system that uses a regular solution model for liquid is fit fairly well by the experimental data, including the

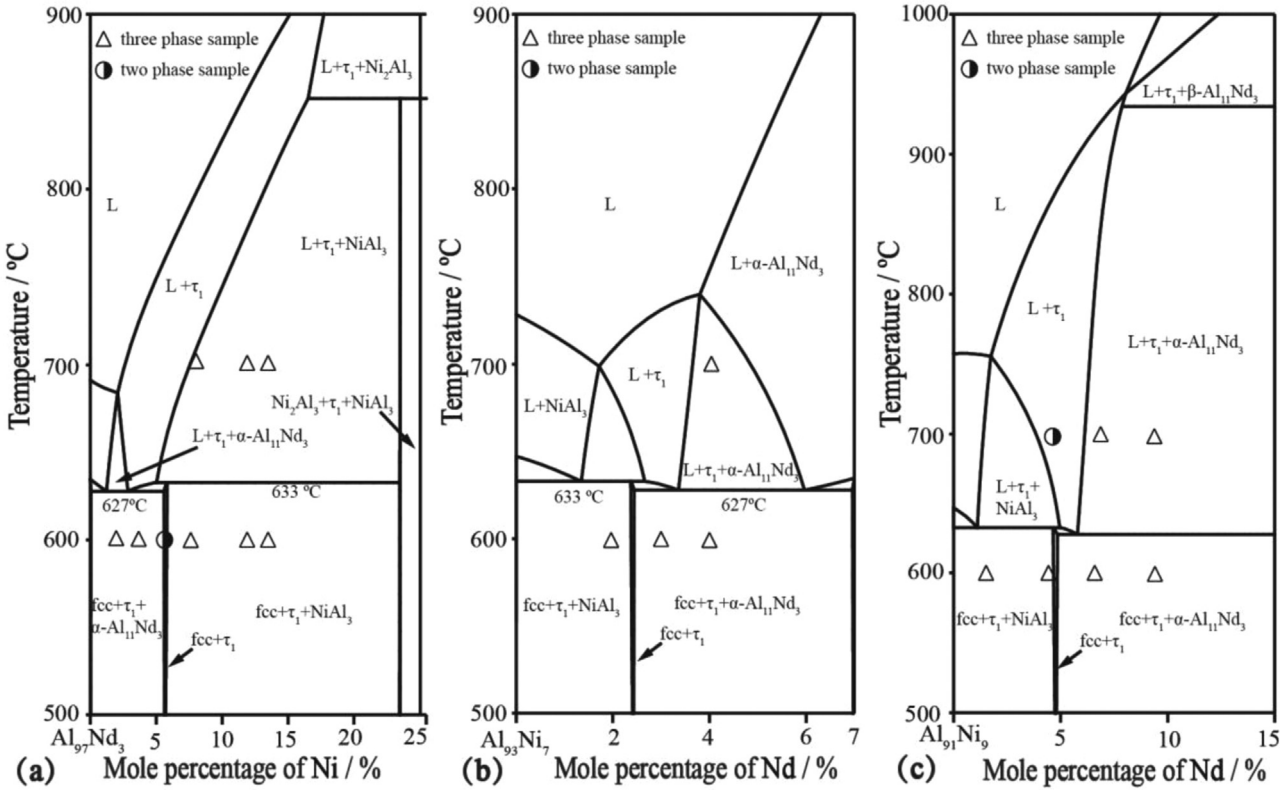

Fig. 6. Calculated vertical Section 3 at% Nd (a), 93 at% Al (b) and 9 at% Ni (c) compared with the experimental data [19].

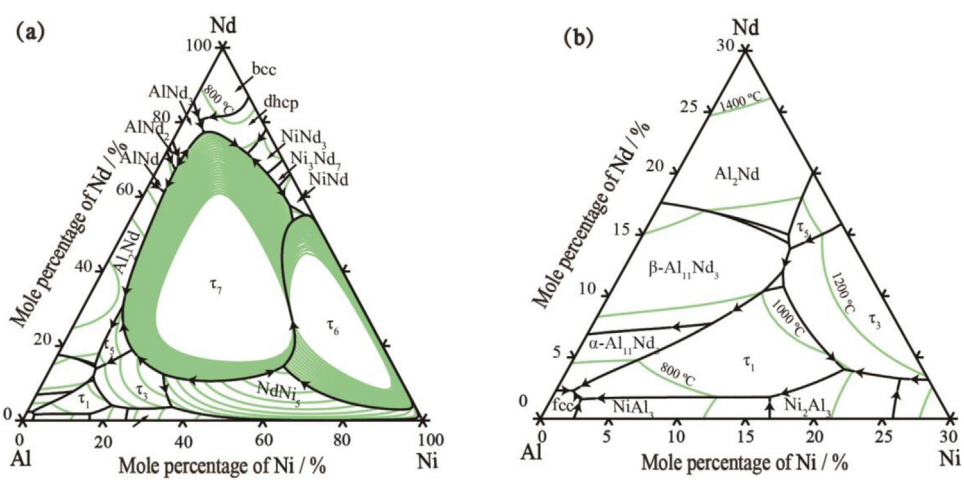

Fig. 7. Calculated liquids projection of the Ni–Al–Nd ternary system. (a) Entire diagram, (b) magnified Al-rich corner.

solubility of Nd in Al. The Al–Nd phase diagram is more accessible. This work can be used as a constitutional system that provides a thermodynamic database for multi-component Ni-based alloys.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.calphad.2018.11.001.

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